

1. Administrative & Study Identification

Official Study Title: A Phase 2, Multicenter, Double-blind, Extension Study to Evaluate the Effects of Sotatercept for the Treatment of Combined Postcapillary and Precapillary Pulmonary Hypertension (Cpc-PH) Due to Heart Failure With Preserved Ejection Fraction (HFpEF) **Public Title:** Extension Study of Sotatercept in People With Pulmonary Hypertension (MK-7962-023) **Short Title/Acronym:** HARMONIZE (MK-7962-023) **Protocol Number:** MK-7962-023 (Other IDs: 7962-023, 2024-515773-99) **NCT Number:** NCT06814145 **Posting ID:** 318369212856179960 **Sponsor/Company Name:** Merck Sharp & Dohme LLC (Merck) **Investigational Product:** Sotatercept (Winrevair / ACE-011) **Phase of Development:** Phase 2 **Trial Status:** RECRUITING **Medical Monitor:** Merck Sharp & Dohme LLC **Clinical Operations Lead**

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the long-term safety and tolerability of sotatercept in participants with combined postcapillary and precapillary pulmonary hypertension (Cpc-PH) due to heart failure with preserved ejection fraction (HFpEF) over an extended treatment period. **Secondary Objectives:** To evaluate sustained hemodynamic, functional, and clinical effects (such as the 6-minute walk distance [6MWD] and pulmonary vascular resistance [PVR]) resulting from prolonged sotatercept administration.

2.2 Background & Rationale Researchers are looking for new ways to treat people with combined postcapillary and precapillary pulmonary hypertension (Cpc-PH) caused by heart failure with preserved ejection fraction (HFpEF). Dysregulation of the transforming growth factor- β (TGF- β) superfamily signaling plays a central role in the pathobiology of pulmonary hypertension. This is an extension study (HARMONIZE) for patients who took part in a parent study (CADENCE / MK-7962-007) evaluating sotatercept for Cpc-PH. The study aims to follow participants' health for a longer duration to learn about the long-term safety of the therapy and if people tolerate it over a longer period of time.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Extension Study) **Control Type:** Double-blind **Blinding:** Double-blind **Randomization:** Continued from the parent study (CADENCE)

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$): 130
Number of Sites: Up to 110 locations globally (including sites in the US, Germany, Poland, Israel, and Belgium). **Estimated Study Start:** 15-Apr-2025 **Estimated Primary Completion:** 29-Mar-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Completed the respective parent clinical study (MK-7962-007 / CADENCE) (Visit 9 and end of treatment [EOT] visit) without discontinuing study intervention and is able to safely enroll into MK-7962-007 (HARMONIZE). 2. Willing to adhere to the study visit schedule and comply with all protocol requirements. 3. Has the ability to understand and provide documented informed consent.

4.2 Exclusion Criteria 1. Did not participate in the sotatercept PAH parent study. 2. Diagnosis of pulmonary hypertension (PH) in World Health Organization (WHO) Group 1, WHO Group 3, WHO Group 4, or WHO Group 5. 3. Chronic liver disease with severe hepatic impairment and/or cirrhosis (e.g., hepatic encephalopathy). 4. Anticipation of more than 1 valve replacement or repair (mechanical or biomechanical) and/or have undergone more than 1 valve replacement or repair. 5. Has severe tricuspid regurgitation due to primary valvular disease (e.g., from endocarditis, carcinoid, or mechanical destruction). 6. Anticipated or undergone heart transplant or ventricular assist device implantation. 7. Has had a study intervention interruption (missed more than the equivalent of 4 consecutive doses between the end of the parent study and the start of this extension study). 8. Has had prior exposure to luspatercept. 9. Pregnant or breastfeeding.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Sotatercept 0.3 mg/kg OR 0.7 mg/kg administered once every 3 weeks (Q3W). **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** Up to approximately 168 weeks or until discontinuation.

5.2 Concomitant & Prohibited Therapy Permitted: Stable background PAH/heart failure therapy continued from the parent study. **Prohibited:** Prior exposure to luspatercept; any study intervention interruptions exceeding the protocol-allowed limit.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening/Roll-over	Day 0 Informed Consent, Roll-over confirmation from parent study
Baseline	Day 1		

First Extension Dose, Baseline Safety Labs | | **Interim Visits** | Q3W | Subcutaneous injection of Sotatercept, Routine Safety Labs | | **Efficacy Assessments** | Weeks 48 & 126 | Functional assessments (e.g., 6MWD test), Hemodynamic Tracking (PVR) | | **End of Study** | Week 168 | Final Vital Signs, IP Accountability, Exit Interview |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Number of Participants Who Experience an Adverse Event (AE):** An AE is any untoward medical occurrence in a clinical study participant, temporally associated with the use of study intervention, whether or not considered related to the study intervention. * **Number of Participants Who Discontinue Study Treatment Due to an AE:** Tracking untoward medical occurrences temporally associated with the study intervention that lead to discontinuation.

7.2 Secondary & Exploratory Endpoints * Change From Baseline in Pulmonary Vascular Resistance (PVR) at Week 48. * Change From Baseline in the 6-Minute Walk Distance (6MWD) at Week 48. * Change From Baseline in the 6-Minute Walk Distance (6MWD) at Week 126. * Sustained hemodynamic and clinical effects in the Cpc-PH population.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous AE monitoring via investigator assessment during Q3W visits. **Serious Adverse Events (SAEs):** Reported within 24 hours of investigator awareness throughout the 168-week extension. **Data Monitoring Committee (DMC):** External independent DMC established by the sponsor to review long-term safety data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who receive at least one dose of the study medication during the extension phase). **Sample Size Justification:** No formal power calculation for the extension; targeted enrollment is approximately 130 participants dictated by the rollover rate of participants successfully completing the CADENCE parent study. **Handling of Missing Data:** Missing long-term functional data will be handled using appropriate imputation methods or analyzed as observed.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standard onsite and remote monitoring to ensure protocol compliance and accurate AE reporting. **Audit Trail:** eCRF locking and comprehensive data clarification mechanisms will be strictly maintained.

11. Study Identifiers & Governance

Primary ID: MK-7962-023 **Registry Number:** NCT06814145 **Posting ID:** 318369212856179960 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC (Merck) **Study Title:** HARMONIZE Extension Study **Official Regulatory Title:** A Phase 2, Multicenter, Double-blind, Extension Study to Evaluate the Effects of Sotatercept for the Treatment of Combined Postcapillary and Precapillary Pulmonary Hypertension (Cpc-PH) Due to Heart Failure With Preserved Ejection Fraction (HFpEF)

12. Clinical Framework (Pulmonary Hypertension Specific)

Disease Areas: Hypertension, Pulmonary **Primary Condition:** Combined Postcapillary and Precapillary Pulmonary Hypertension (Cpc-PH) **Name:** Hypertension **Preferred Name:** Hypertensive disease **Preferred UMLS Name:** Pulmonary Hypertension **Semantic Type:** Disease or Syndrome **Definition:** The presence of chronic increased pressure in the systemic arterial system. [<https://orcid.org/0000-0002-0736-9199>] **Diagnosis Threshold:** Cpc-PH due to Heart Failure with Preserved Ejection Fraction (HFpEF) diagnosed in parent study **Medical Methodology:** Activin signaling inhibitor / TGF- β superfamily ligand trap (Sotatercept) + Background PH/HF therapy **Terminology Codes:** * **MeSH Code:** D006973 * **HPO Code:** HP: 0000822 * **SNOMED ID:** 38341003 * **SNOMED Hierarchy:** 49601007|Disorder of cardiovascular system|Disorder of cardiovascular system (disorder), 64572001|Disease|Disease (disorder)

13. Study Procedures & Methodology

Management Pathway: Continuous Q3W Subcutaneous (SC) Injection Pathway **Education Protocol (HCP):** Subcutaneous injection administration and specialized AE tracking for TGF- β inhibitors. **Education Protocol (Patient):** Site care for Q3W subcutaneous injections and prompt reporting of epistaxis or telangiectasias. **Quality Audit Mechanism:** HCP adherence tracking to Q3W dosing schedules and safety reporting timelines.

14. Observation & Follow-Up Schedule

Total Duration: Up to 168 weeks **Onsite Visit Cadence:** Every 3 weeks (Q3W) for dosing **Remote Monitoring Frequency:** Supplemental to onsite visits as required by the investigator **Checklist Review Interval:** Major efficacy reviews at Week 48 and Week 126

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					